

Functions and transformations - Set 2

TMUA practice questions by topic. Specification MM1.7, MM8.2, MM8.3, MM8.4. 6 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Which of the following functions, each with domain the set of all real numbers, is one-to-one?
- A $f(x) = x^3 - x$
 - B $f(x) = x^2 + 2x$
 - C $f(x) = \cos x$
 - D $f(x) = x^3 + x$
 - E $f(x) = |x|$
- 2 The function f satisfies $f(x) = x^2$ for all x . Which of the following transformations maps $y = f(x)$ onto $y = 4x^2 - 8x + 4$?
- A a translation of 1 unit in the positive x -direction followed by a stretch parallel to the y -axis with factor 4
 - B a translation of 2 units in the positive x -direction followed by a stretch parallel to the y -axis with factor 2
 - C a translation of 1 unit in the negative x -direction followed by a stretch parallel to the y -axis with factor 4
 - D a stretch parallel to the y -axis with factor 4 followed by a translation of 4 units in the positive y -direction
 - E a stretch parallel to the x -axis with factor 2 followed by a translation of 2 units in the positive x -direction

- 3 $f(x) = x^2 + 1$. Which of the following statements is/are true?
- I $f(f(x)) = x^4 + 2x^2 + 2$ for all x
 - II $f(x + 1) = f(x) + 1$ for all x
 - III $f(2x) = 4f(x) - 3$ for all x
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 4 Let $f(x) = \sqrt{x - 3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?
- I $g(f(x)) = x$ for all $x \geq 3$
 - II $f(g(x)) = x$ for all real x
 - III $f(g(x)) = |x|$ for all real x
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 5 The function f is defined by $f(x) = |x - 2| + |x + 2|$ for all real x . What is the minimum value of $f(x)$, and for how many values of x is it attained?
- A minimum 0, attained for exactly two x
 - B minimum 2, attained for exactly one x
 - C minimum 4, attained for exactly two x
 - D minimum 4, attained for infinitely many x
 - E minimum 4, attained for exactly one x

- 6** Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

I $g(x) = (x + 2)^2$

II $g(x) = (x - 2)^2$

III The graph of g has its vertex at $(-2, 0)$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III